Here's the complete step-by-step guide to get everything running from scratch:

🔧 Step 1 — Wire up your hardware

Connect everything to your Arduino Uno like this:

Component	Pin on component	Arduino pin
DHT22	VCC	5V
DHT22	GND	GND
DHT22	DATA	Digital Pin 3
MQ135	VCC	5V
MQ135	GND	GND
MQ135	AOUT	A0
LED	Anode (+)	Digital Pin 8 (via 220Ω resistor)
LED	Cathode (–)	GND
Buzzer	+	Digital Pin 9
Buzzer	–	GND
LCD (I2C)	VCC	5V
LCD (I2C)	GND	GND
LCD (I2C)	SDA	A4
LCD (I2C)	SCL	A5

💻 Step 2 — Upload the Arduino sketch

1. Connect Arduino to your laptop via USB cable
2. Open **Arduino IDE**
3. Open your smart_study_desk.ino file
4. Go to **Tools → Board** → select **Arduino Uno**

5. Go to **Tools → Port** → select the port that says Arduino (e.g. COM3 on Windows, /dev/ttyACM0 on Linux/Mac)
6. Click the **Upload** button (the right arrow ➡️ at the top)
7. Wait for it to say **"Done uploading"** at the bottom

✅ Your LCD should light up and show **"Smart StudyDesk — Initializing.."** for 2 seconds, then start showing temperature and humidity.

🐍 Step 3 — Set up Python environment

Open a terminal / command prompt and run these commands one by one:

bash

```
pip install opencv-python mediapipe ultralytics pyserial
```

This will take a few minutes the first time. Wait for all of them to finish installing.

⚙️ Step 4 — Set the correct COM port in Python script

1. Open smart_study_desk.py in any text editor
2. Find this line near the top:

python

```
arduino = serial.Serial(port='COM3', baudrate=9600, timeout=1)
```

3. Change COM3 to match your actual port:
   - **Windows** → open Device Manager → Ports (COM & LPT) → look for Arduino
   - **Mac** → it will be something like /dev/cu.usbmodem14201
   - **Linux** → it will be something like /dev/ttyACM0

▶️ Step 5 — Run the Python script

⚠️ **Important:** Close Arduino IDE's Serial Monitor first if it's open. Python and Serial Monitor can't use the same port at the same time.

Now in your terminal, navigate to the folder where your Python file is saved and run:

bash

```
python smart_study_desk.py
```

👀 Step 6 — What you should see

On your laptop screen:

- A webcam window opens titled **"Smart Study Desk"**
- A green skeleton is drawn over your body (MediaPipe landmarks)
- A status bar at the top shows one of:
  - 🟢 Good - Keep studying!
  - 🔴 BAD POSTURE - Sit straight!
  - 🟠 PHONE DETECTED - Stay focused!

On the LCD:

- Alternates every 3 seconds between:
  - Temp: 24.5 C / Humidity: 42%
  - Gas level: 312 / Air: Good
- When an alert happens, LCD shows the alert message instantly

Buzzer + LED:

- Bad posture → 2 short beeps + LED on
- Phone detected → 1 long beep + LED on
- Bad air/temp → 3 rapid beeps + LED on
- All good → silent, LED off

❌ To stop the program

- Click on the webcam window and press **Q** on your keyboard
- Everything shuts down cleanly — webcam releases, Arduino clears alerts, serial port closes

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🛠️ Common problems and fixes

Problem	Fix
Port already in use error	Close Arduino IDE Serial Monitor before running Python
LCD is on but shows blank boxes	Change 0x27 to 0x3F in the .ino file and re-upload
ModuleNotFoundError in Python	Run pip install again for the missing library
Webcam window doesn't open	Change cv2.VideoCapture(0) to cv2.VideoCapture(1) in the Python script
Arduino not detected	Try a different USB cable — some cables are charge-only and don't carry data
YOLO model not loading	Let it finish downloading (~6MB) on first run, needs internet